

Basic Principles of Medicine 2
Module: Digestion and Metabolism
Lecture 22

Enteric Motility

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Note: The lecture slides that are available for download and print-out contain complementary slides that are not presented during the lecture.

These slides have a blue background and contain useful information about the material discussed during the lecture.

Recommended Reading

Rhoades, R. & Bell, D. (2023); *Medical Physiology: Principles for Clinical Medicine*, 6th Edition; Wolters Kluwer. Chapter 25

[Gastrointestinal Motility | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library](#)

Lecture Objectives I

SOM.MK.II.BPM2.3.DM.1.PHYS.1101. Discuss the esophageal pressure in terms of its normal resting range at different sites, method of measurements and variations with the respiratory cycle.

SOM.MK.II.BPM2.3.DM.1.PHYS.1102. Compare and contrast the structure and function of the upper versus lower esophageal sphincters.

SOM.MK.II.BPM2.3.DM.1.PHYS.1144. Explain the role of upper and lower esophageal sphincters in swallowing.

SOM.MK.II.BPM2.3.DM.1.PHYS.1103. Illustrate the dynamic pressure changes that occur in the esophagus after initiation of the swallowing reflex.

SOM.MK.II.BPM2.3.DM.1.PHYS.1104 Describe how dysfunction in the spatial or temporal characteristics of the esophageal pressure wave and/or sphincter relaxation can lead to swallowing defects and disorders such as heart burn, achalasia and aspiration of food.

SOM.MK.II.BPM2.3.DM.1.PHYS.1105. Describe the anatomical locations and role of interstitial cells of Cajal as slow wave pacemakers and mediators of inputs from the enteric nervous system.

SOM.MK.II.BPM2.3.DM.1.PHYS.1106. Explain the control of peristalsis by the enteric nervous system.

SOM.MK.II.BPM2.3.DM.1.PHYS.1107. Describe how extrinsic nerves (sympathetic and parasympathetic) affect GI motor patterns.

SOM.MK.II.BPM2.3.DM.1.PHYS.1108. Compare and contrast the functional roles of intrinsic and extrinsic enteric innervation in GI motor control.

SOM.MK.II.BPM2.3.DM.1.PHYS.1109. Describe major motor patterns in the GI tract and their functions during fasting (migrating motor complex or MMC) and during digestion.

Lecture Objectives II

SOM.MK.II.BPM2.3.DM.1.PHYS.1110. List the afferent neuro-muscular pathways involved in the swallowing reflex, and the 2 major nuclei in the brain stem that integrate these motor pathways.

SOM.MK.II.BPM2.3.DM.1.PHYS.1111. Describe the storage, digestion, and motility roles of the stomach.

SOM.MK.II.BPM2.3.DM.1.PHYS.1112. Describe how distension of organs affects GI reflexes and alters responses to other regulatory inputs.

SOM.MK.II.BPM2.3.DM.1.PHYS.1113. Illustrate how luminal pressure and luminal stretch initiate motor reflexes in GI organs.

SOM.MK.II.BPM2.3.DM.1.PHYS.1114. Describe how these inputs (luminal pressure and luminal stretch) are integrated by intrinsic and extrinsic neural pathways including enteric ganglia, prevertebral ganglia, spinal cord and brain.

SOM.MK.II.BPM2.3.DM.1.PHYS.1115. Describe the function of colonic motility, in mediating formation of haustra and haustral shuttling, mass movements through the transverse and distal colon, and defecation.

SOM.MK.II.BPM2.3.DM.1.PHYS.1116. Describe the sequence of events in the colon and anal sphincters occurring during reflexive defecation, differentiating those movements under voluntary control and those under autonomic control.

Lecture Outline

- GI motility patterns
 - Tonic/phasic, peristalsis, segmentation
 - Fasting versus fed state
- Mechanisms and regulation of GI motility
 - Neural input, slow waves and ICC cells
- Motility in specific GIT segments
 - Esophagus, stomach, small and large intestines
 - Motility patterns, function, regulation
- GIT reflexes & defecation

Part 1

GI motility patterns

GI smooth muscle and motility

- Depolarization of smooth muscle → contraction
 - circular muscle contraction → lumen diameter decrease
 - longitudinal muscle contraction → tract length decrease
- Tonic contractions
 - sustained for minutes to hours
 - sphincter specialised muscle (lower esophageal, pyloric, ileocecal and internal anal)
- Phasic contractions
 - sustained for seconds before return to relaxation
 - from esophagus – rectum except in stomach fundus

Key motor patterns & functions

1) Segmental contractions (nonpropulsive)



↑ mixing
and
churning



↑ digestion
and
absorption

2) Peristaltic Contractions



Propulsion in caudal direction

3) Reservoir Function



Enhanced by presence of sphincters. Delayed peristaltic contractions also allow stomach and large intestines to act as reservoirs.

Definitions

- **Motility:** the action of the muscles of the GI tract that mix and propel its contents from mouth to anus.
- **Peristalsis:** the involuntary constriction and relaxation of the muscles of the intestine or another canal, creating wavelike movements that push the contents of the canal forward.
- **Segmentation:** contractions are a common type of mixing motility seen especially in the small intestine - segmental rings of contraction chop and mix the ingesta. Alternating contraction and relaxation of the longitudinal muscle in the wall of the gut also provides effective mixing of its contents.

Major GIT motility states

Interdigestive (fasting)
state

MMC (migrating motor complex)

- “cleans” stomach and gut during fasting

Fed/digestive state

Segmentation & peristalsis

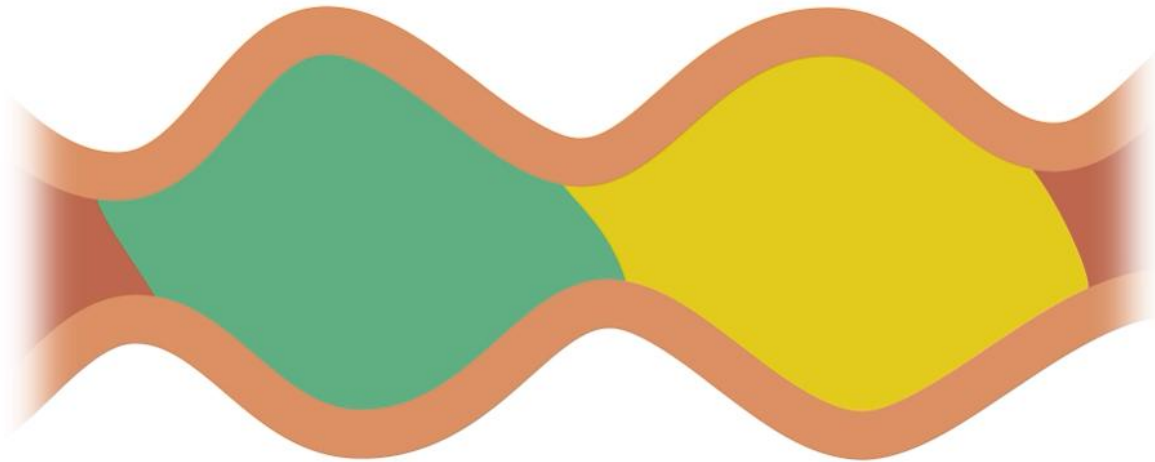
- Segmentation aids mixing
- Peristalsis moves “food” along GIT

Fed state motility I: Segmentation

Key function: Mixing luminal content. NO forward movement.

Contraction: Small section of small intestine contracts, splitting chyme and sending it in both directions.

Relaxation: Section of SI now relaxes, allowing bolus of chyme to come back together.

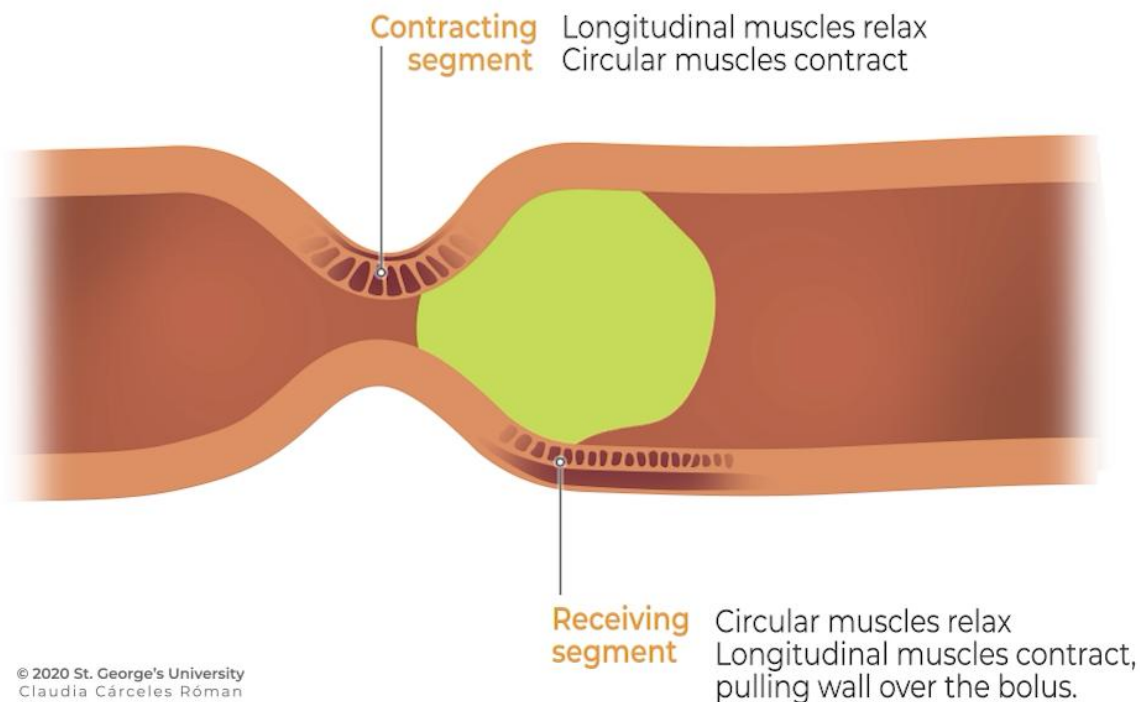


Fed state motility II: Peristalsis

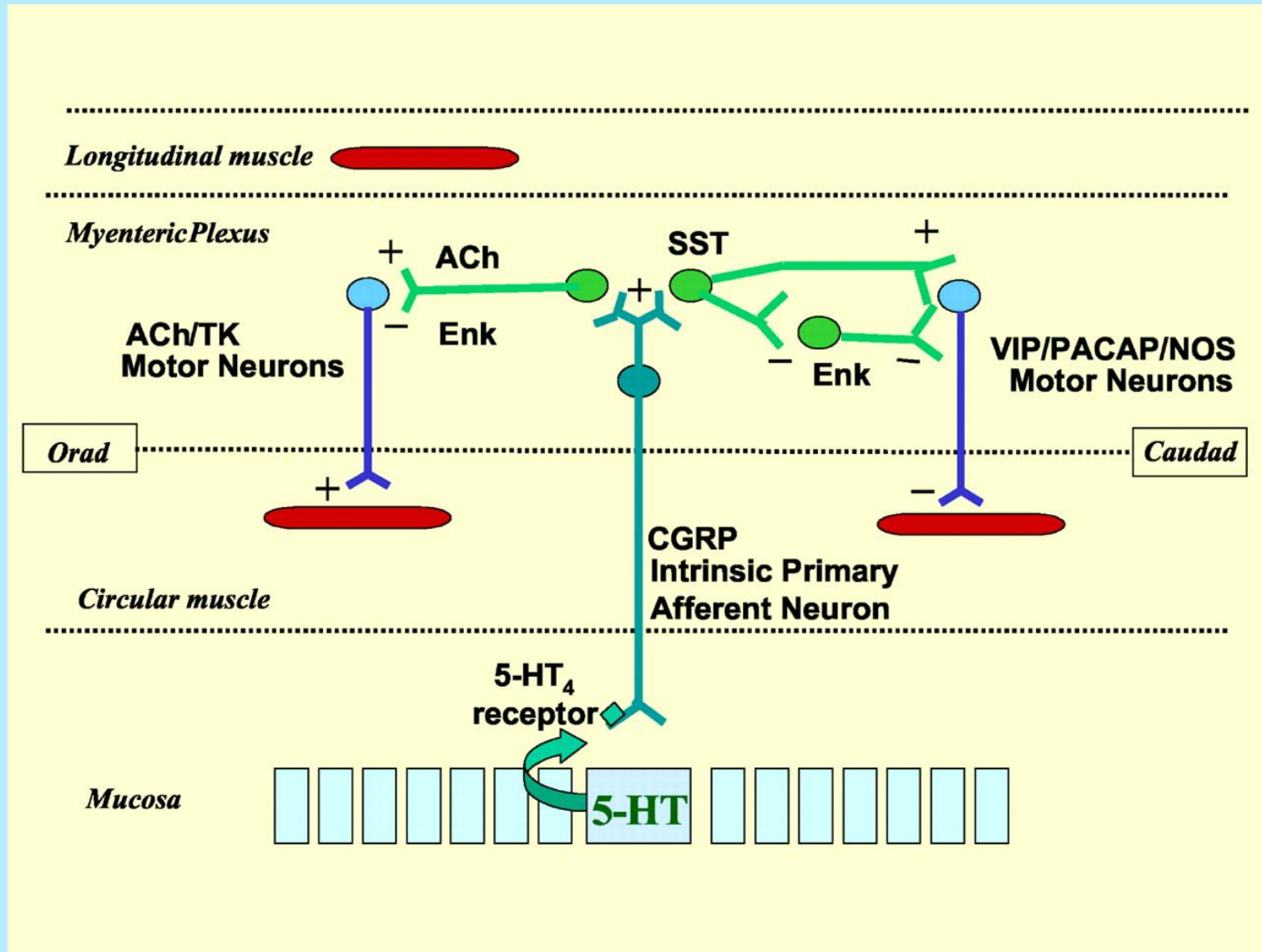
The peristaltic reflex is a carefully coordinated sequence of contractions and relaxations in the circular and longitudinal muscles which moves luminal content caudally.

Most (~80%) peristaltic waves travel < 2 cm, aiding mixing. Others travel > 50 cm, chiefly moving luminal content down the GIT.

Neurotransmitters involved in the **oral** (behind the bolus or chyme) contraction are ACh and Substance P, those involved in the **caudal** (in front of) relaxation are VIP and nitric oxide.



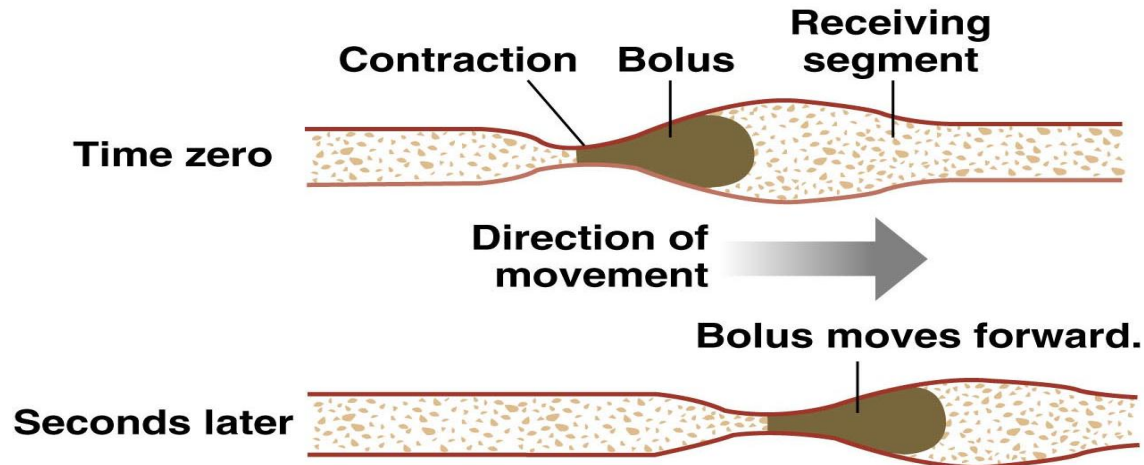
Intestinal peristaltic reflex details



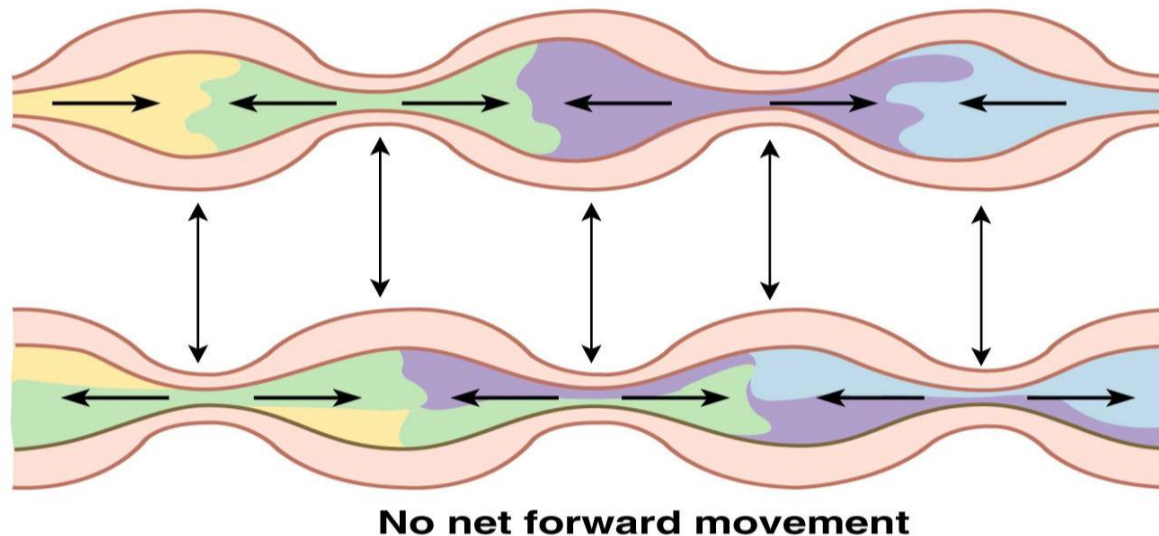
John R. Grider J Pharmacol Exp Ther 2003;307:460-467

Summary slide: Phasic contractions

(a) Peristaltic contractions create forward movement.

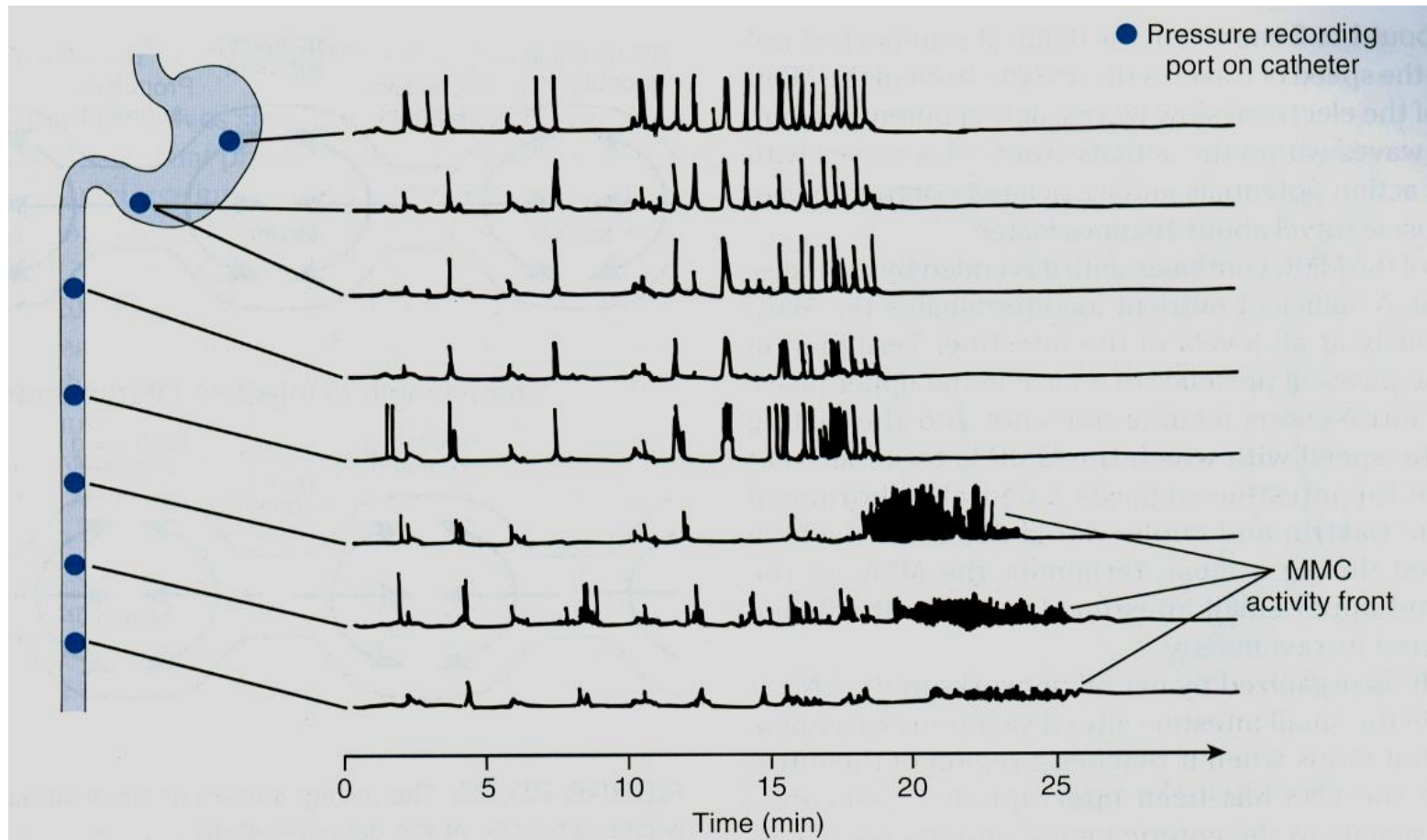


(b) Segmental contractions are responsible for mixing.



The Migrating Motor Complex (MMC)

- Characteristic pattern of activity that occurs during fasting.
- Strong propulsive contractions that pass from stomach through SI
- Frequency roughly every 90 min.
- Disappears in severe gut motility dysfunction



Characteristics of the MMC

- Functional roles
 - Preventing bacterial overgrowth in small intestine
 - Propulsion of undigested material
- Neural basis
 - Observed in transplanted bowel, hence the MMC does not depend on extrinsic (ANS) input
 - Generated intrinsically by the ENS
 - MMC “clock” (90 min) also generated by ENS
- Onset and termination
 - **Motilin** induces the MMC
 - MMCs stop after ingestion of a meal
 - MMCs start again during the next fasting period

Part II

Mechanisms and regulation of GI motility

Motility regulation overview

- GI tract produces contractions and motility patterns intrinsically
- Motility patterns are coordinated by
 - Local ENS innervation (excitatory and inhibitory motor neurones and interneurons)
 - Interstitial cells embedded in smooth muscle layers (ICC & PDGFR cells)
- ANS alters contractile force but not pattern
- Stretch, hormones and paracrine factors can alter both contractile force and motility pattern

Integration of GIT motility regulation

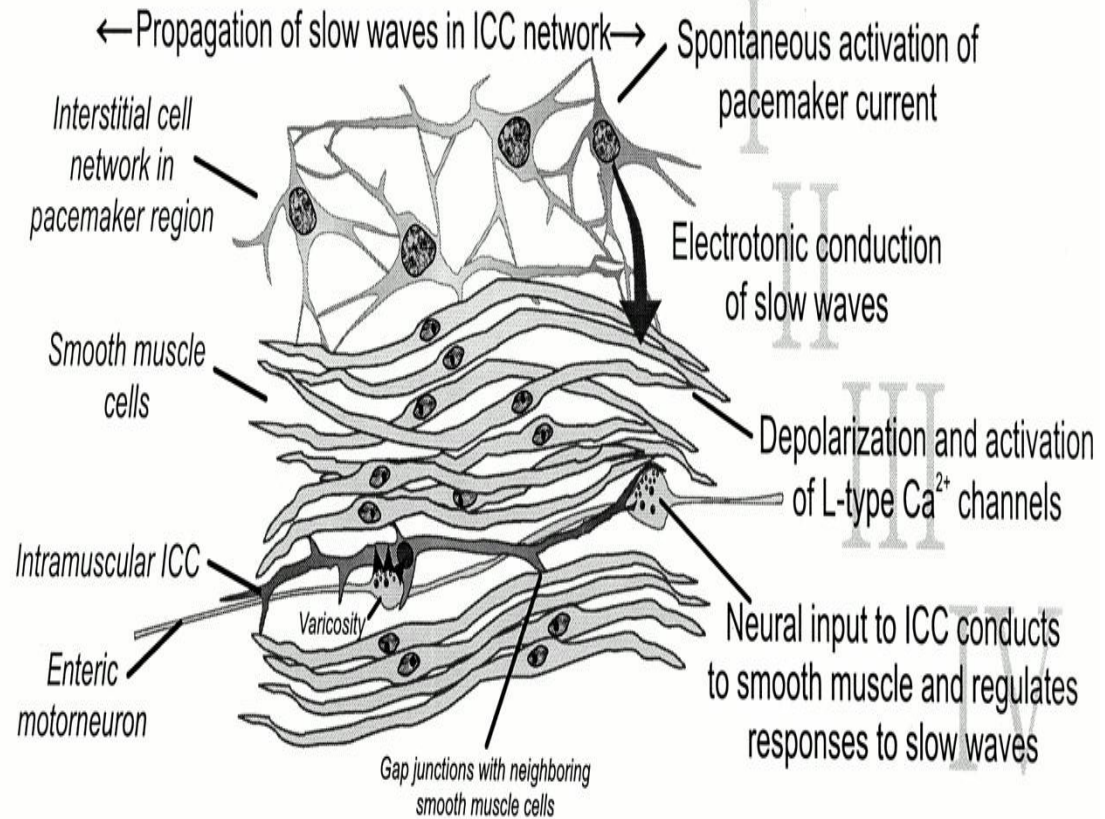
- ANS (extrinsic input), ENS (intrinsic), stretch and hormonal signals are integrated
- Parasympathetic (Ach): ↑ rate of generation of action potentials → increased motility
- VIP, NO: ↓ rate of generation of action potentials → decreased motility
- luminal content → stretch → increased motility
 - via local mechano- and osmoreceptors linked to endocrine and paracrine mediators
 - also direct electrical effects of stretch

Interstitial Cells of Cajal (ICC)

- located between circular and longitudinal muscles and within each muscle layer (different subtypes)
- ICC and SM cells form an 'electrical syncytium'
 - electrically coupled by gap junctions to each other, the ENS and neighbouring SM cells, enabling motor coordination
- intrinsic GI pacemakers by generating Slow Waves
 - cyclical opening of Ca^{2+} channels (depolarization) and opening of K^{+} channels (repolarization)
- integrate gut mechanics (stretch) into motility pattern
 - ICCs are stretch-sensitive
 - luminal content $\rightarrow$ stretch $\rightarrow$ depolarization
 - slow wave peak + stretch $\rightarrow$ contractions
 - coordination of peristalsis and segmental contractions

Slow waves

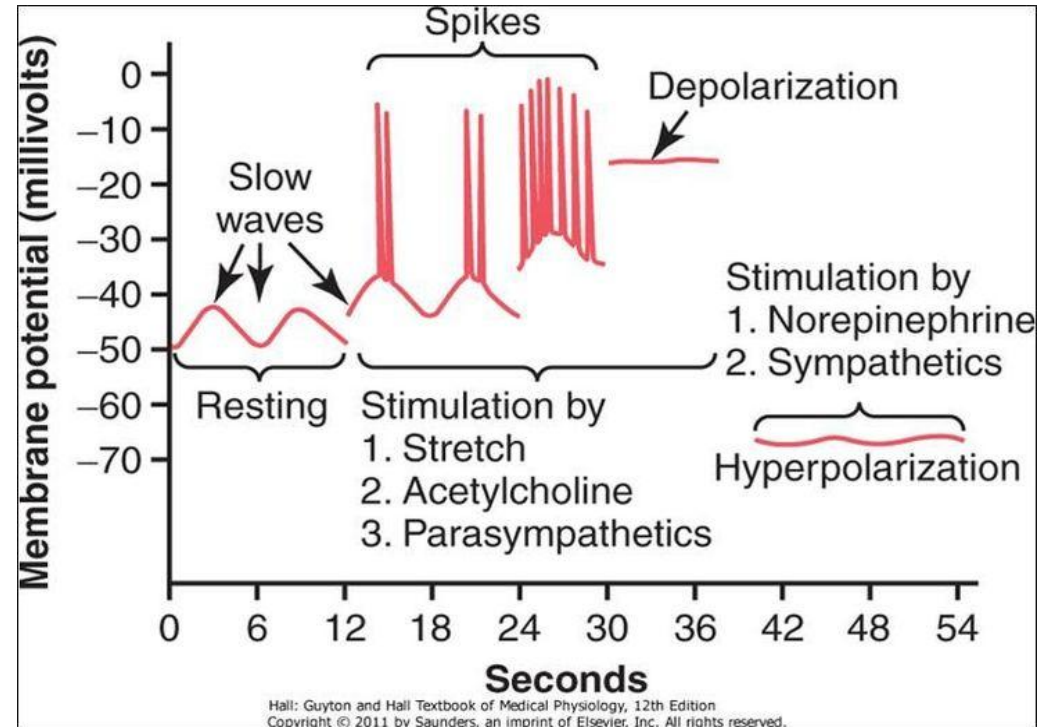
- sub-threshold oscillating membrane potentials
- *not* action potentials
- occur spontaneously in sections of GIT
- originate in the interstitial cells of Cajal (ICC)
- drive basic electrical rhythm (BER)



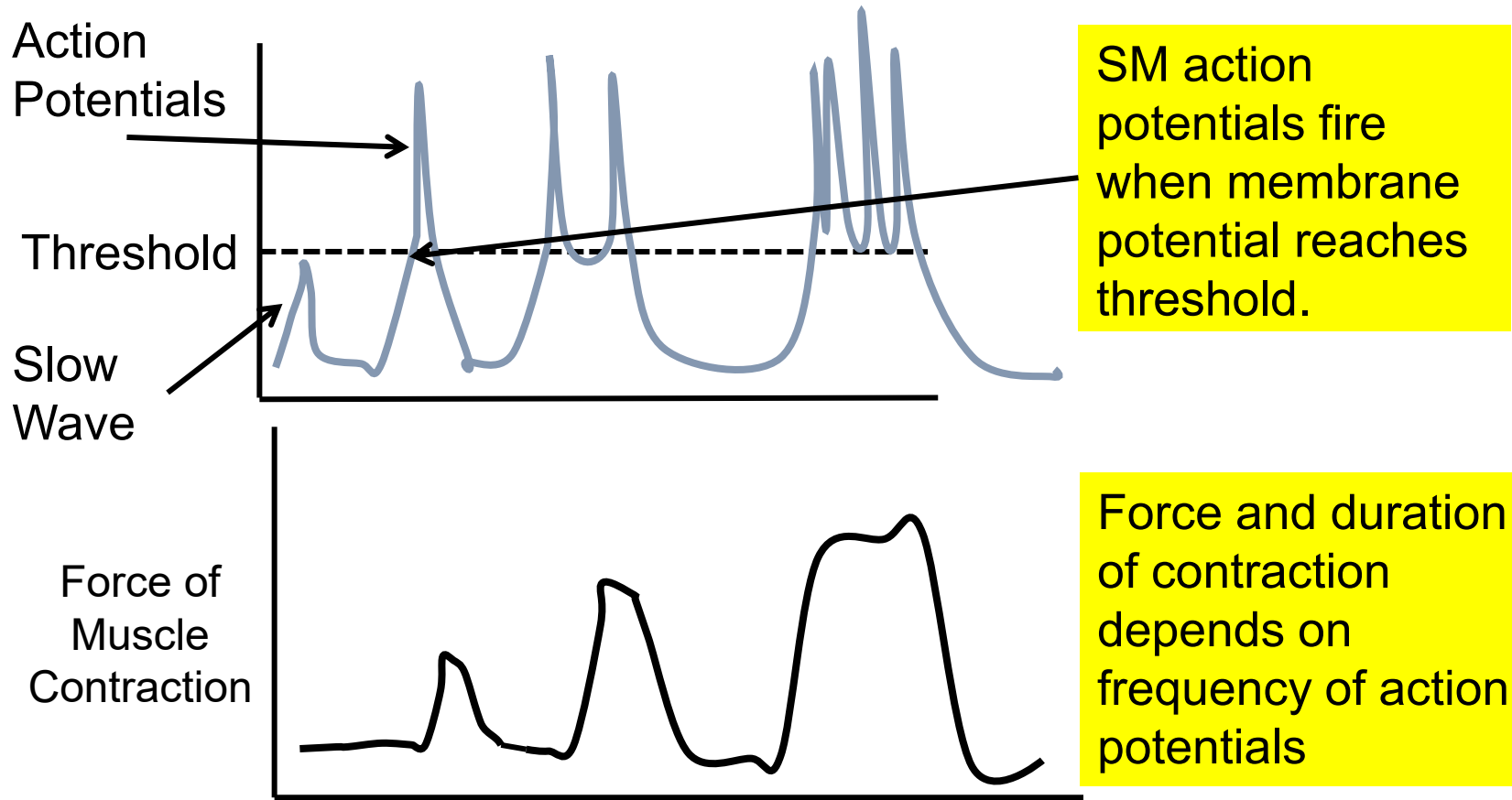
<http://physiologyonline.physiology.org/content/nips/15/6/291/F2.large.jpg>

Slow waves and action potentials

- ICC and SM cells electrically connected via gap junctions
- SM cell action potentials on top of slow waves initiate contractions
- Slow wave frequency not influenced by neural or hormonal input
- But frequency of the action potentials on top of the slow waves is modified by neural and hormonal influences



Slow waves, APs and contractions



Slow wave reaches threshold → Ca channels open → Ca enters cell
→ Action Potential is generated → Smooth Muscle contracts

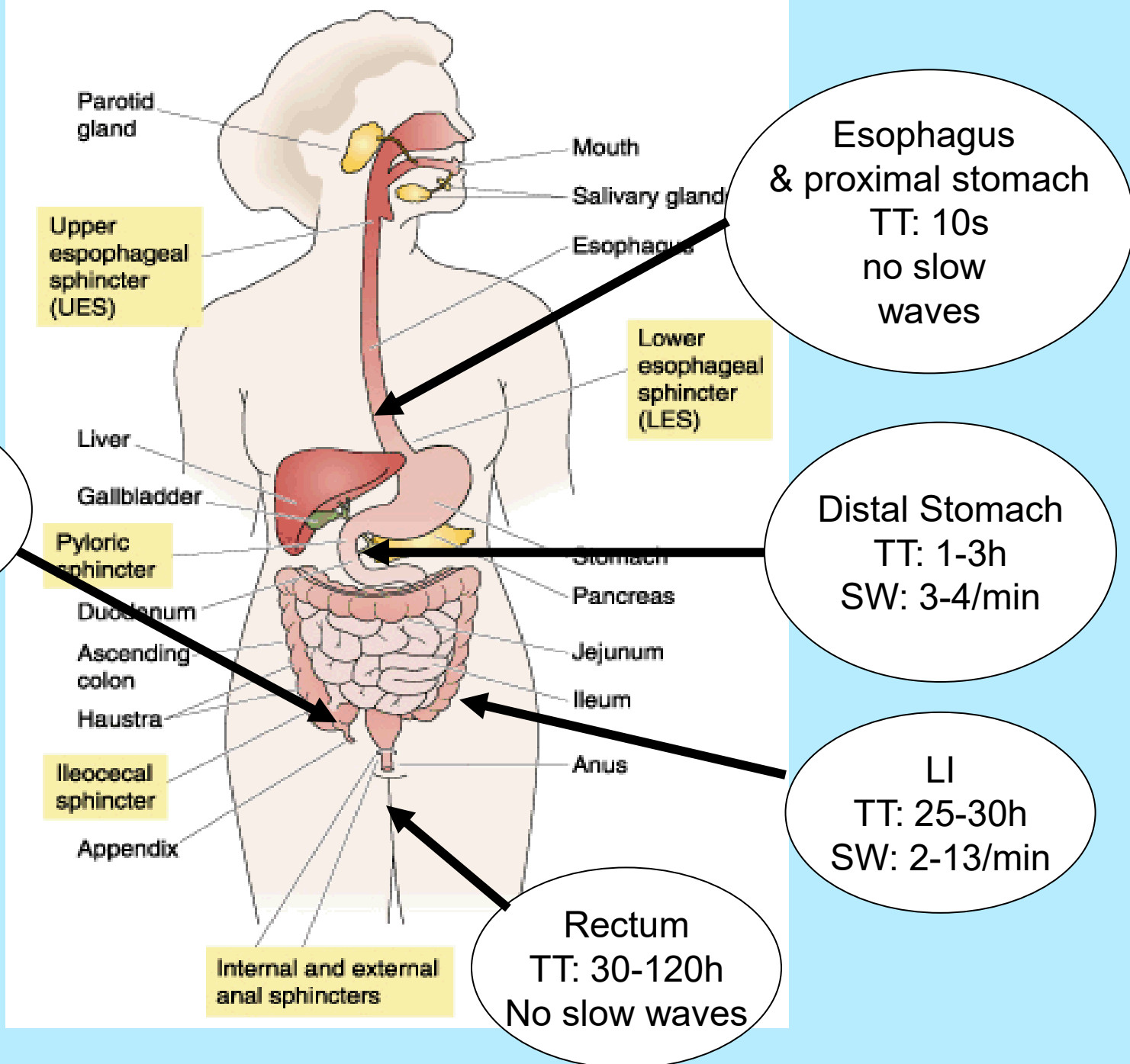
Slow wave summary slide

- When a slow wave reaches threshold, voltage-gated Ca channels open
- Action Potentials are generated by the Ca influx, causing smooth Muscle to contract
- Force of contraction is regulated by the amount of Ca that enters the muscle cell
- Inward Ca fluxes are followed by opening of Ca-activated K channels, repolarising the cell
- Frequency of muscle contraction is set by the rate of slow wave generation
- BER is essential for phasic contractions of the GIT

Slow waves and transit time

- Transit time varies along the GI tract, but is constant and characteristic for each part
- Slow waves set the maximum frequency of contractions for each part of the GI tract
- Typical SW frequencies:
 - No slow waves in proximal stomach (storage) and esophagus
 - Distal stomach: 3-4 / min
 - SI: 11-12/ min
 - LI: 2-13 / min

Transit times



Part III

Motility in specific GIT segments

Clinical assessment of motility

- Manometry
 - a flexible tube (catheter) with pressure sensors is passed through nose into esophagus and stomach
 - also wireless pills (pressure, pH, temperature)
- Gastric emptying scintigraphy
 - imaging progression of a radioactively labelled meal
- pH monitoring
 - e.g. in esophagus for gastric reflux

Motility patterns in each segment

KEY

M: motility

Salivary gland

Upper
esophageal
sphincter

Esophagus

Lower
esophageal
sphincter

Liver

Gallbladder

Pyloric valve

Pancreas

Ileocecal
valve

Rectum

Anal
sphincters

ORAL CAVITY AND ESOPHAGUS

M: swallowing, chewing

- Primary and secondary peristalsis
- Role of Sphincters
- Clinical Correlation - GERD (Heart burn), Achalsia

STOMACH

M: peristaltic mixing and propulsion

Fed State

- Receptive relaxation
- Mixing and digestion
- Gastric emptying

Fasting state

- MMCs

SMALL INTESTINE

M: mixing and propulsion primarily by segmentation;

- Segmentation contractions
- Peristaltic contractions
- Gastroileal reflex

LARGE INTESTINE

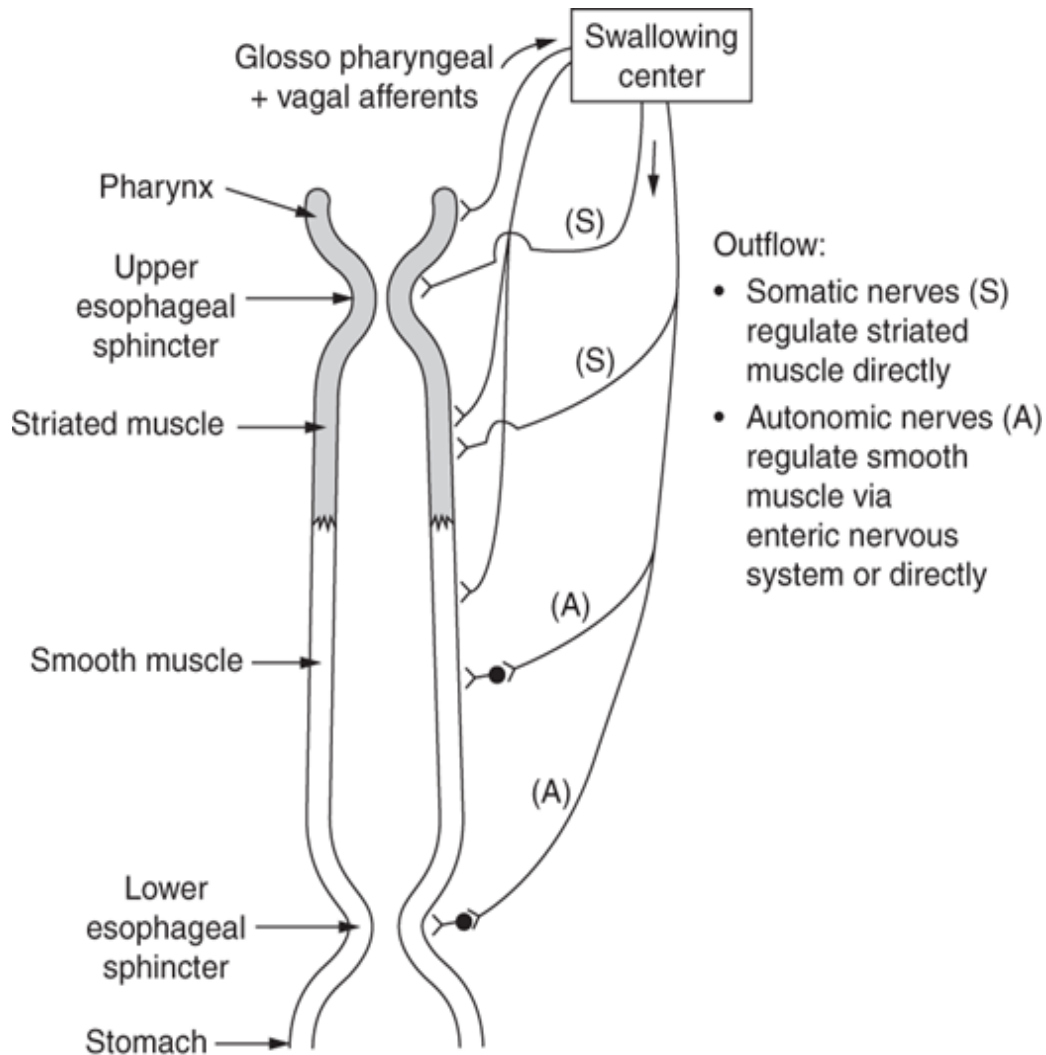
M: segmental mixing; mass movement for propulsion

- Role of Sphincters
- Segmentation, Mass movements
- Defecation
- Reflexes
- Clinical correlation : -
- Constipation/ Diarrhea
- Hirschsprung Disease

The Esophagus

- conveys food and liquid to stomach
- upper 1/3: striated muscle
 - somatic innervation (conscious control)
 - upper esophageal sphincter (UES)
- middle 1/3: transitional
 - Smooth and striated muscle
- lower 1/3: smooth muscle
 - ANS & ENS innervation (no conscious control)
 - Lowest zone (3-4 cm): lower esophageal sphincter (LES)
- zone between UES and LES (body of esophagus) is quiescent at rest

Swallowing Reflex: Innervation



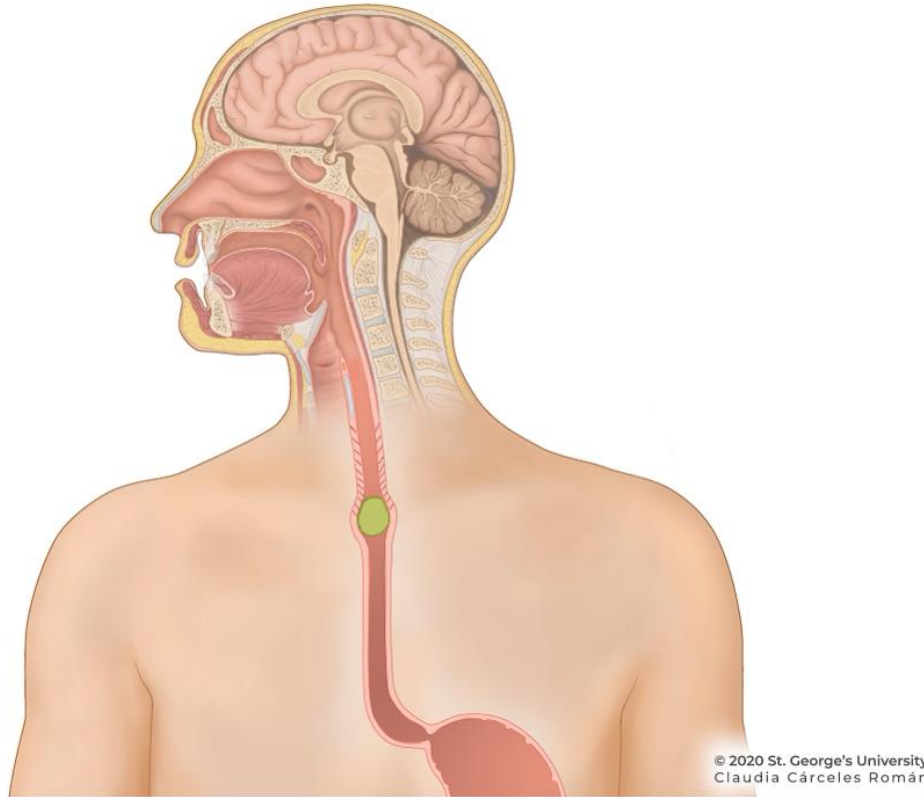
Source: Barrett KE: *Gastrointestinal Physiology, Second Edition*:
www.accessmedicine.com
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Coordinated by CN IX and X via the medulla.

Nucleus ambiguus: somatic nerves - regulate pharynx and striated areas of the esophagus (upper 1/3) via sequential activation.

Dorsal motor nucleus: autonomic nerves – control smooth muscle of esophagus (lower 2/3) via ENS by projecting onto inhibitory and excitatory neurons in the esophageal myenteric plexus.

Swallowing



Primary peristaltic wave: caused by intentional swallowing and coordinated via central innervation.

Secondary peristaltis: caused by esophagus distension, e.g. due to reflux or incomplete swallowing.

1. Voluntary stage
Food is squeezed by tongue up and back against soft palate.

2. Pharyngeal stage
Bolus of food stimulates epithelial swallowing receptors in pharynx to send impulses to the brain stem, and a fast peristaltic wave pushes bolus into the upper esophagus (< 2 sec).

3. Esophageal stage
Primary/secondary peristaltic waves push bolus into stomach (5-8 sec).

Esophageal sphincters

- esophagus pressure *lower* than atmospheric and stomach, hence esophagus must have 2 sphincters
- upper and lower esophageal sphincter (UES and LES)
- both normally contracted

Esophageal manometry

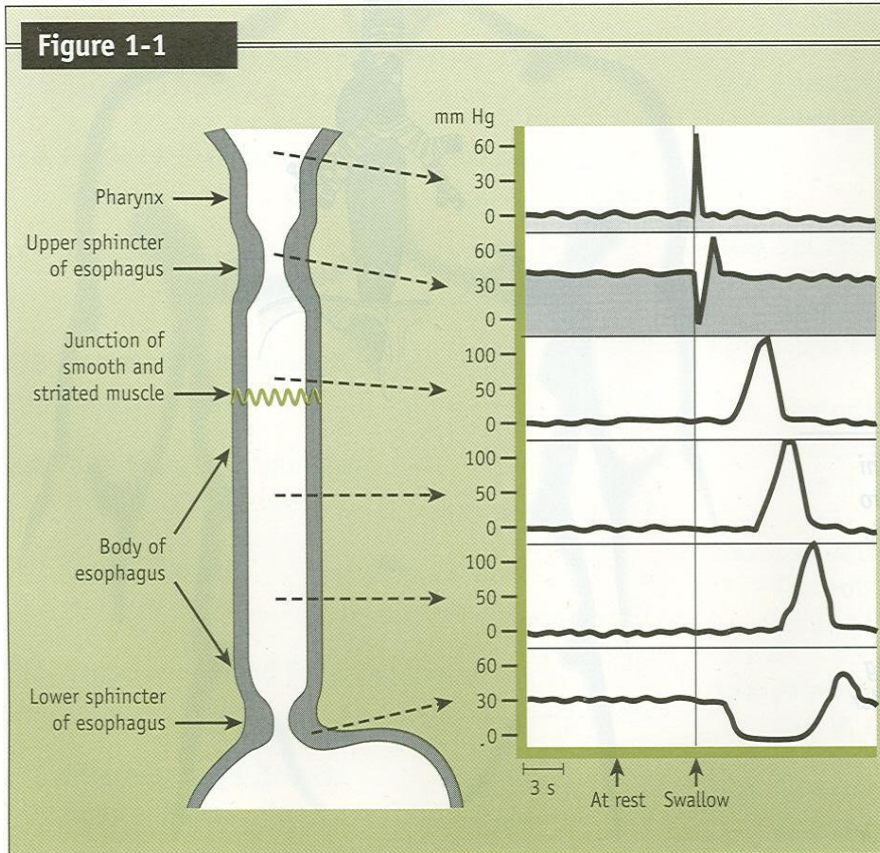


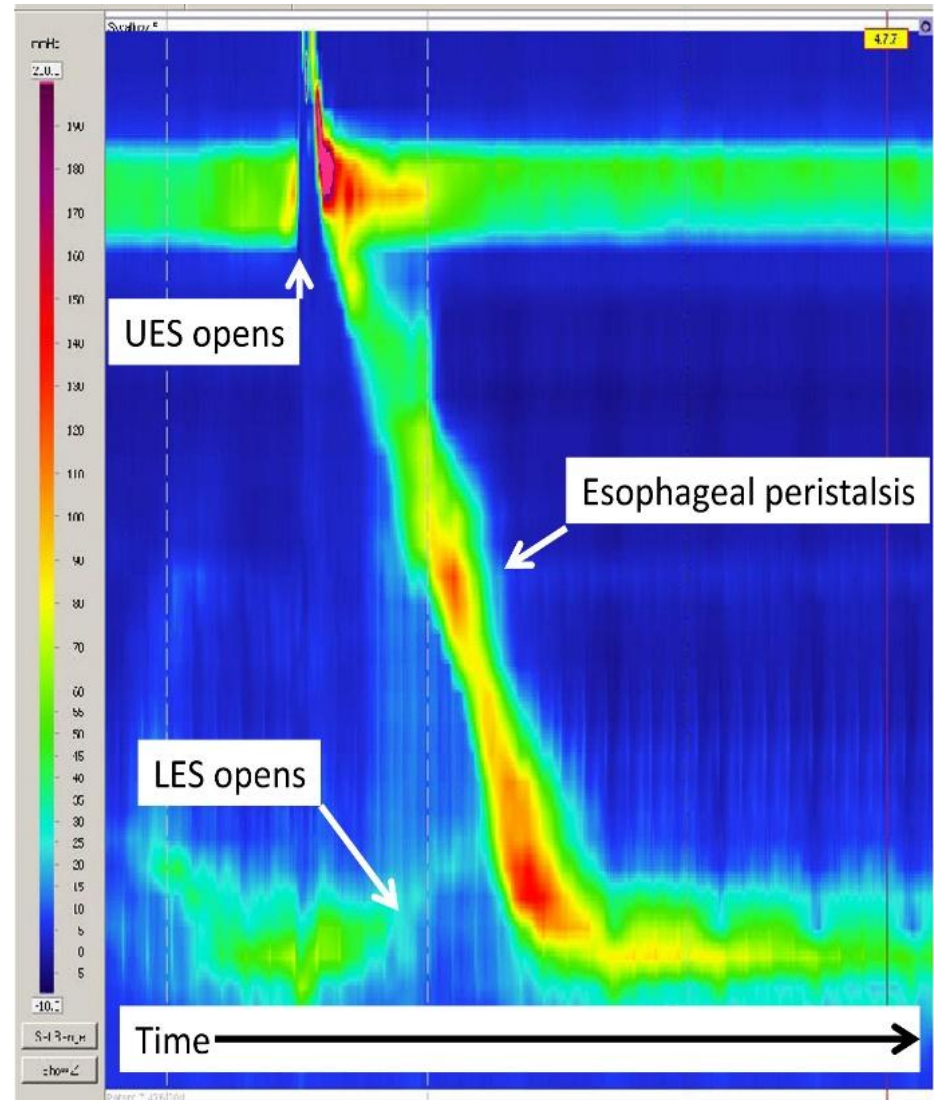
Diagram from Gastroenterology by L Brandt

Esophageal pressures during a swallow

- typical relative pressures
 - UES: + 40 mm Hg
 - mid esophagus: -5 mm Hg (subatmospheric)
 - LES: + 20 mm Hg
 - Stomach: +5 mm Hg
- normal swallowing
 - UES relaxes temporarily
 - primary peristaltic wave sweeps down esophagus
 - LES relaxes transiently to allow the bolus to enter the stomach

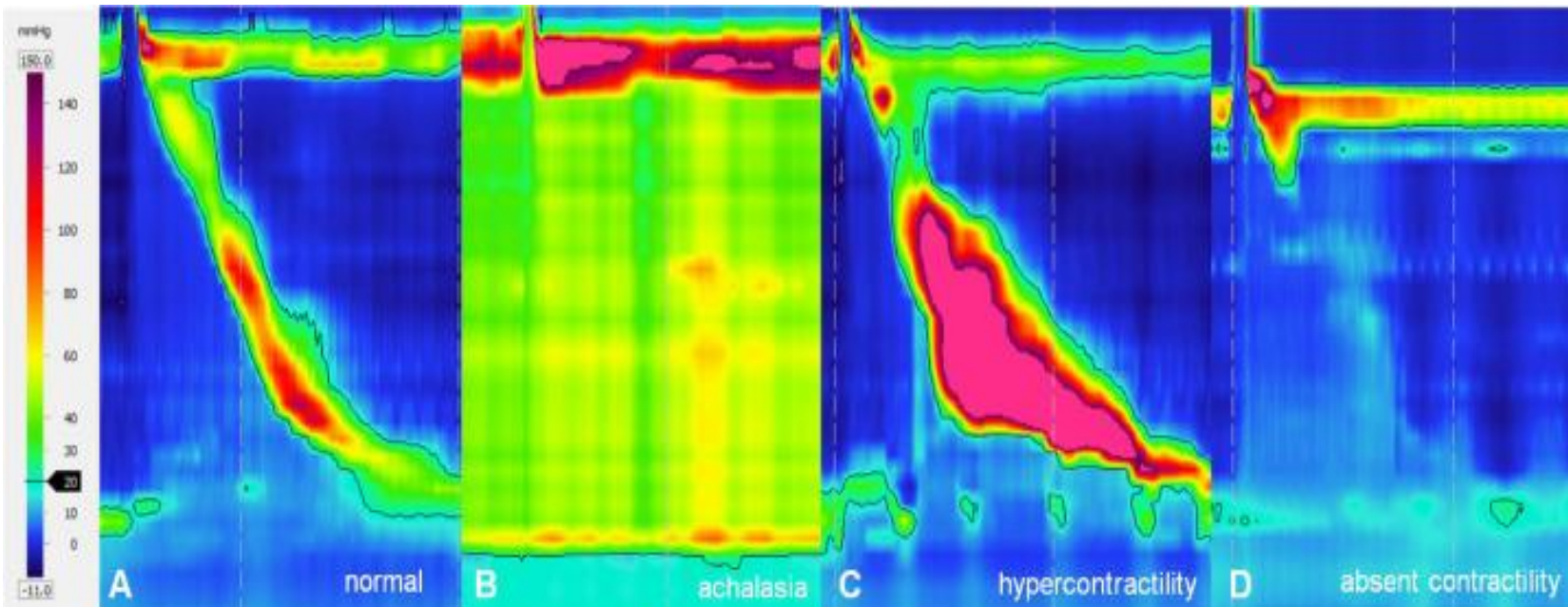
High-resolution manometry

- Detailed spatio-temporal pressure plot
- Pressures at each location are color-coded
- Image illustrates normal swallowing



Clinical correlates

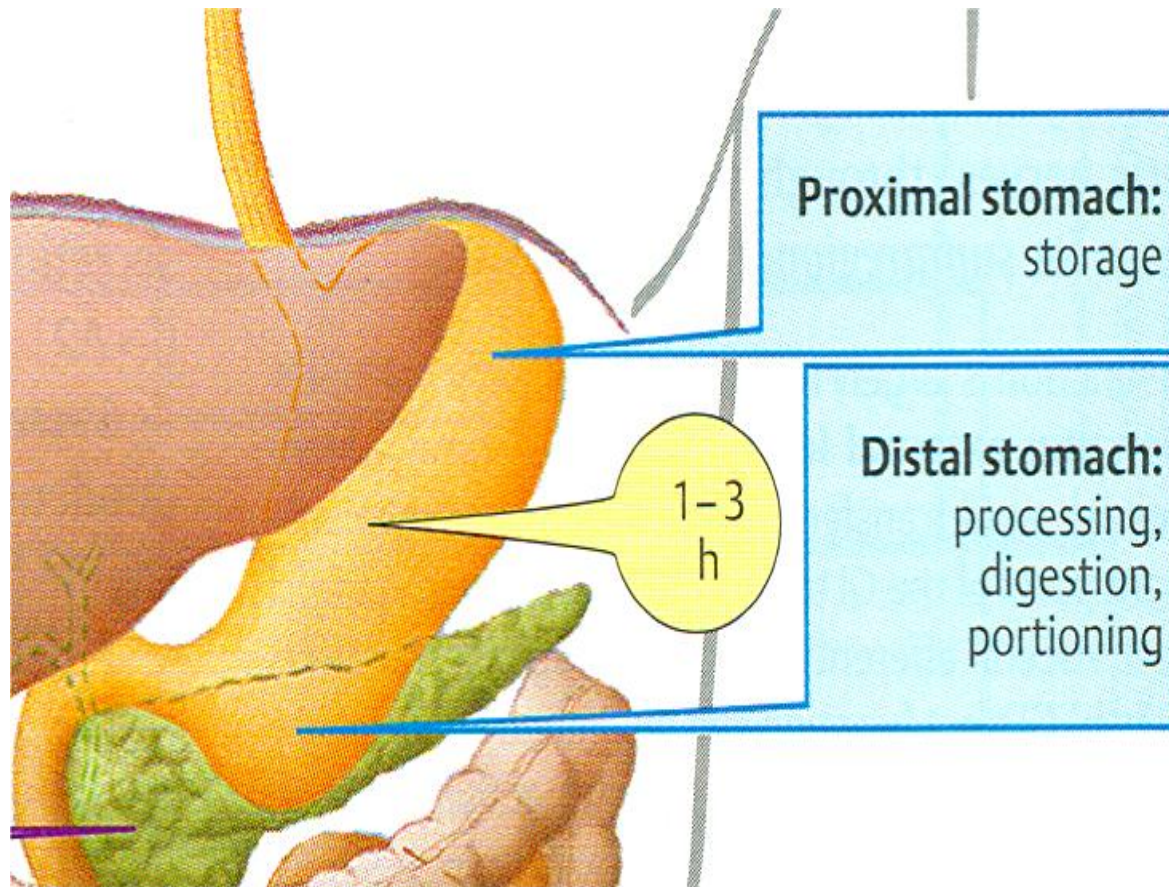
- Acid reflux
 - when stomach content escapes into esophagus via LES
 - normal: acid in esophagus → secondary peristalsis → return to stomach
 - Gastro-Esophageal Reflux Disease (GERD): chronic reflux due to impaired LES function, causing mucosal damage
 - major symptom: pyrosis (heartburn)
- Achalasia (“absence of relaxation”)
 - loss of peristalsis and/or failure of the LES to relax properly
 - symptoms: difficulty swallowing, food aspiration, malnourishment



Achalasia

Pathophysiology	• Degeneration of inhibitory ganglions of the myenteric plexus • Lower Esophageal Sphincter fails to relax during swallowing
Clinical presentation	• Chronic dysphagia to solids and liquids • Regurgitation of food. • Heartburn, weight loss
Diagnosis	• Manometry: incomplete LES relaxation (indicated by failure to show reduction in LES pressure during swallowing), Reduced peristalsis of distal esophagus. • Barium esophagography: smooth "bird-beak" narrowing at gastroesophageal junction.
Management	• Myotomy or pneumatic balloon dilation • Botulinum toxin injection, nitrates and CCB
CCB = calcium channel blocker; LES = lower esophageal sphincter.	

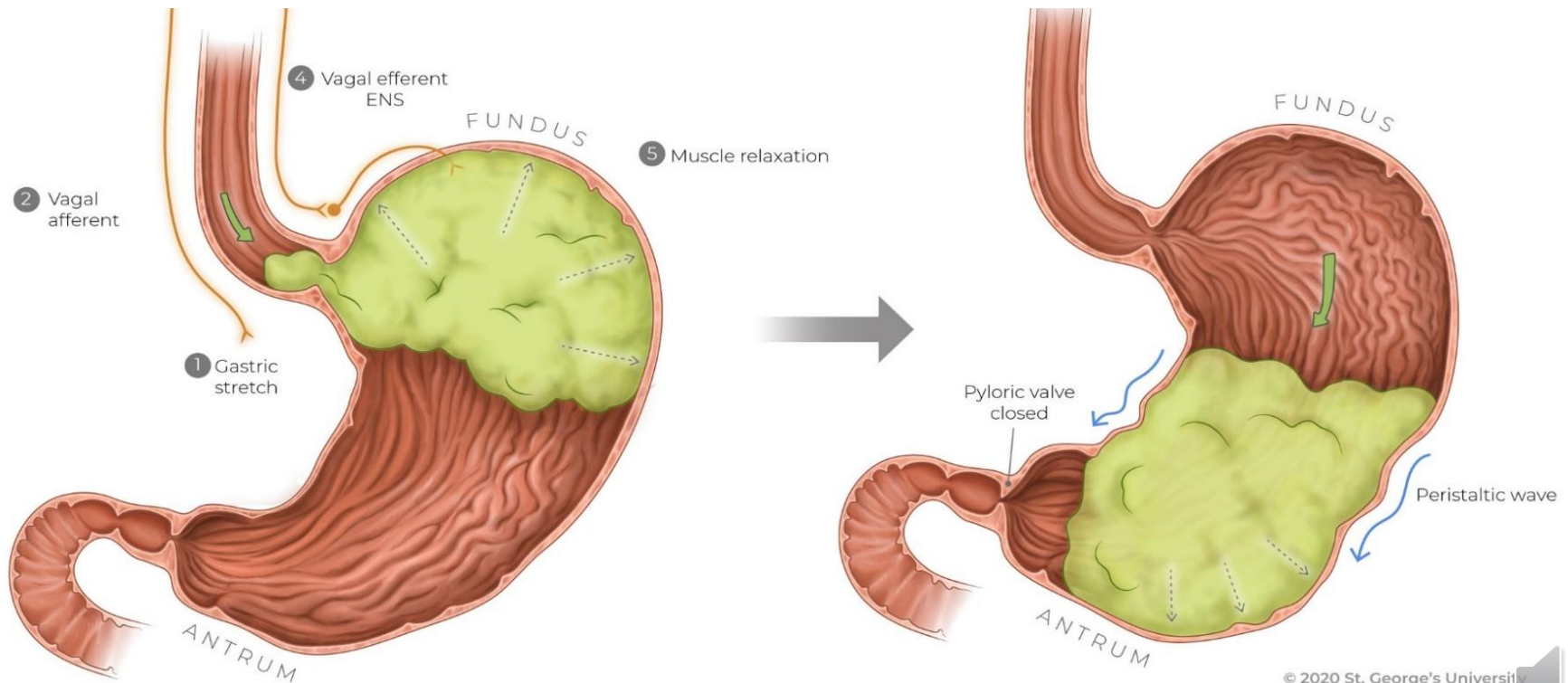
Stomach functional divisions



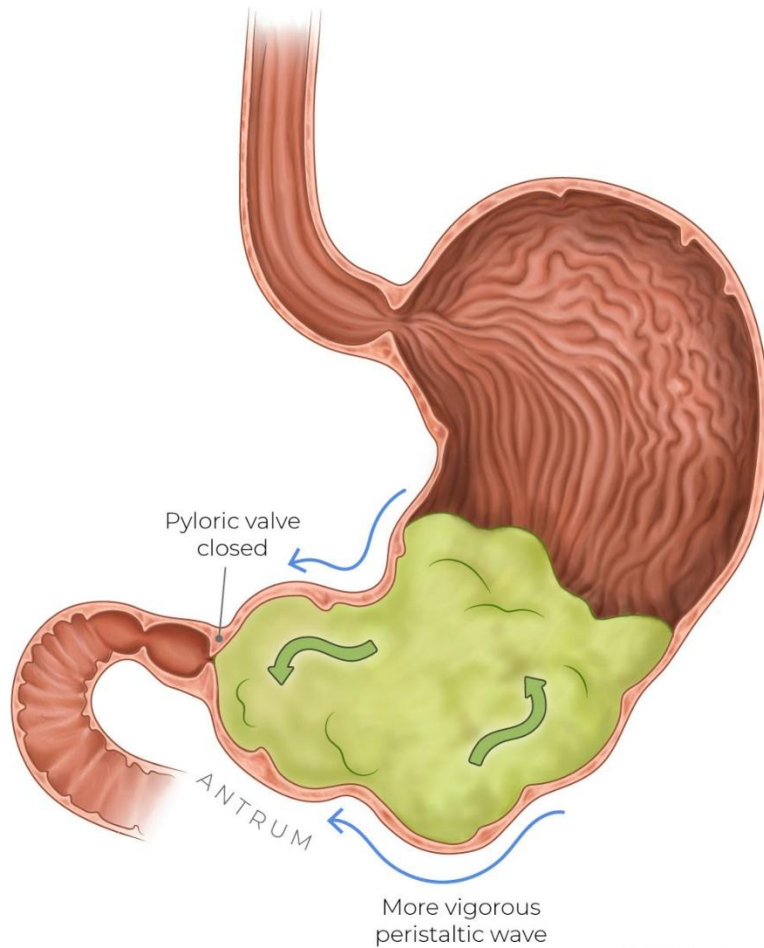
- Tonic contractions
- No slow waves
- Reservoir function
- Phasic contractions
- active slow waves (BER pacemaker)
- Digestive functions
 - mixing
 - emptying

Gastric Motility: reservoir function

Reservoir function: Food arriving at the stomach induces receptive relaxation (vagovagal and via ENS). Gastric pressure remains constant until a threshold of additional volume. Stomach volume may increase from 0.05 to 1.5L.



Gastric Motility: mixing and digestion

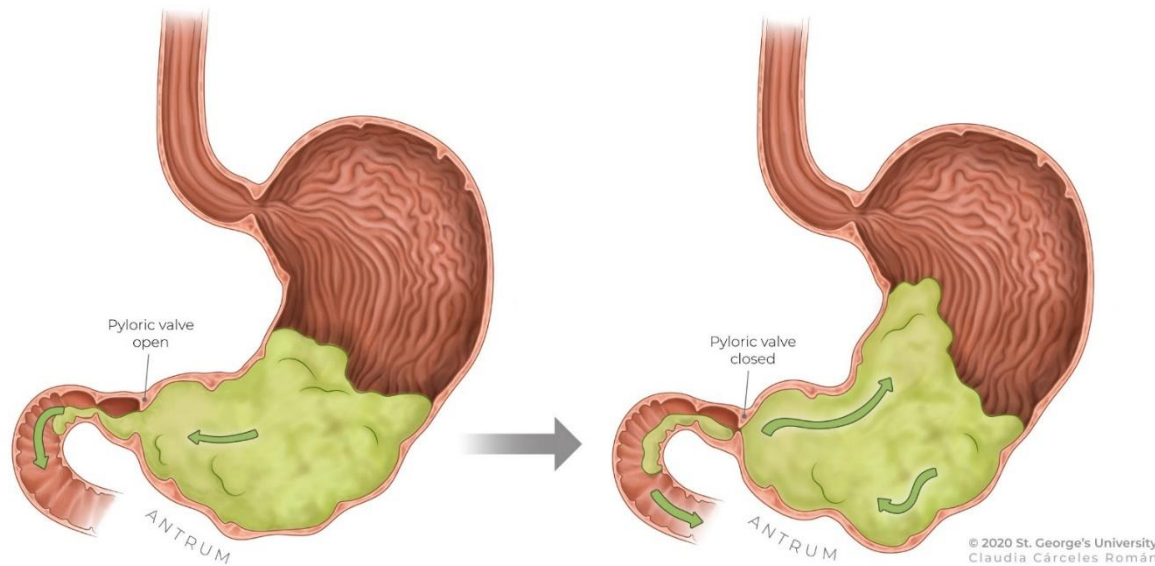


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Weak peristaltic constrictor waves (mixing) start from gastric pacemaker cells in smooth muscle located on greater curvature of the stomach.

These constrictor waves move into the antrum, become more powerful and food trapped near the antrum undergoes grinding.

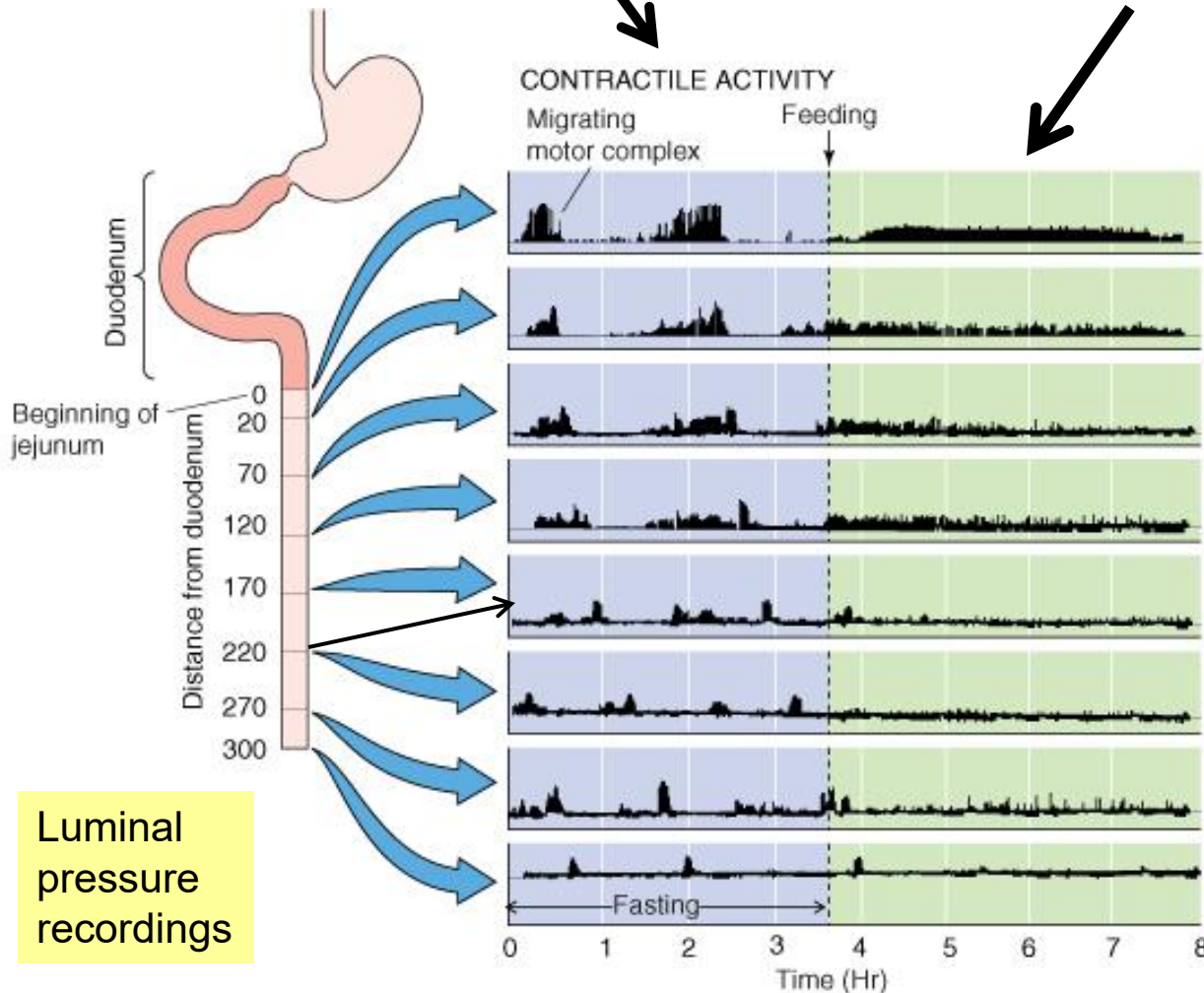
Gastric Motility: emptying



- Particles > 2 mm do not leave stomach in immediate post prandial period
- Rate of emptying: Water $>$ acidic and caloric fluids $>$ fatty food
- Gastric emptying regulated by hormones and ANS (Increased motility by gastrin and vagus nerve, and reduced by CCK, secretin and GIP).
- Pyloric Sphincter
 - antral stomach contraction leads to closure, keeping contents in stomach
 - only small amounts food are transferred to the duodenum
 - retropulsion: most chyme forced back into gastric antrum

Motility: Small Intestine

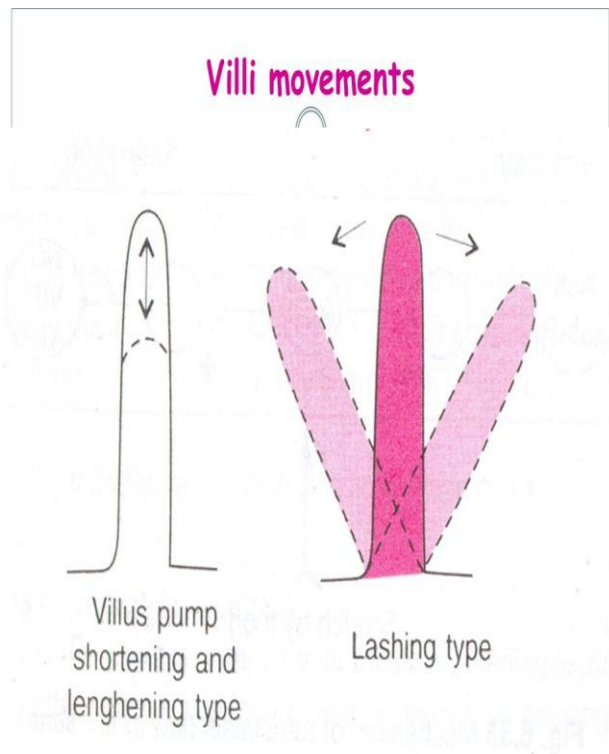
Interdigestive state



Fed / digestive state

Continuous contractile activity throughout the SI whilst food is being digested and absorbed

Small Intestine motility: Villi movements



- Villi are the tiny finger like projections that line the small intestines to increase the surface area for absorption
- Local nervous reflex in response to chyme
- Muscularis mucosa – filament of the smooth muscle fiber
- Lashing type
- Villus pump

Motility: Large Intestine

- **Haustra** – sac-like segments, appear after the contractions of the colon.

Cecum and Proximal colon

- Distension of material in ileum relaxes ileocecal sphincter
- Distention of cecum with fecal material closes sphincter.

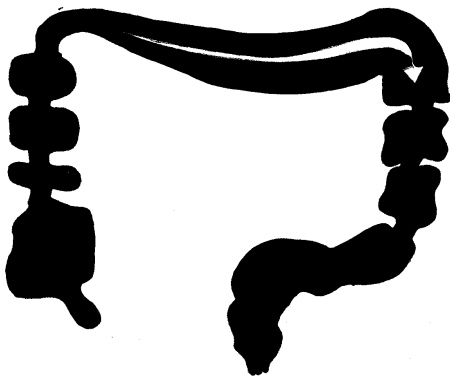
- Haustral shuttling - mixes the contents (segmentation), intrinsic nerves, slow, bacterial growth
- Mass movements – move fecal material distally for long distances, gastrocolic reflex, responds to gastrin and ENS

Distal colon

- Water absorption, fecal material is semisolid, mass movements push it into rectum



“haustral shuttling”



“mass movements”

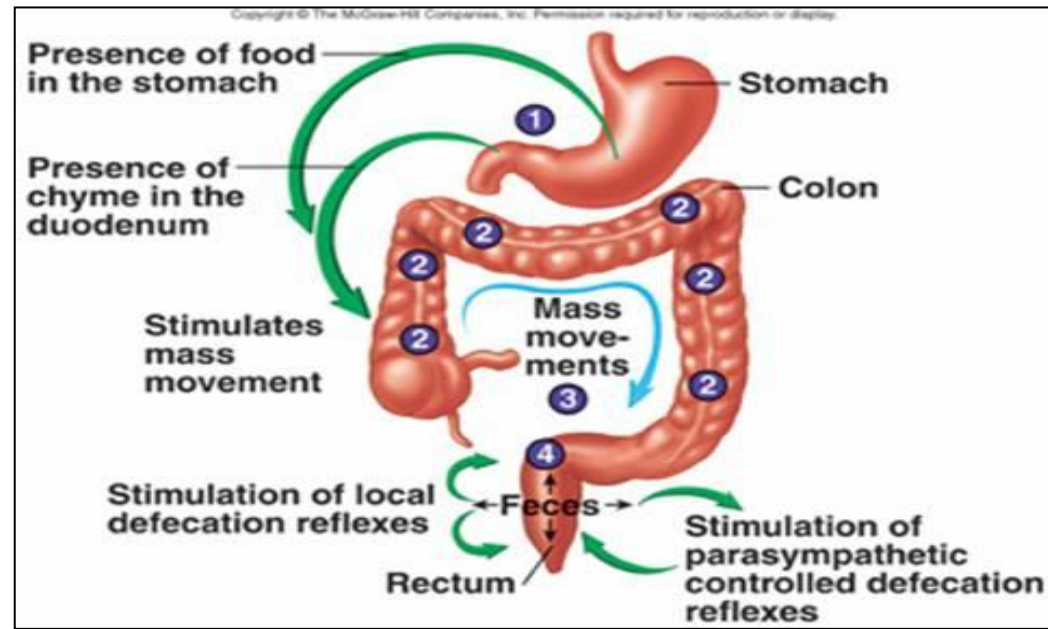
Part IV

GI reflexes

Feedforward vs feedback Reflexes

Feedforward

- Gastrocolic – food in stomach increases frequency of mass movements
- Colocolic/duodenocolic – propels stool caudally by proximal muscle constriction distension and distal dilatation
- Gastroileal – stomach distension leading to increased ileal peristalsis and relaxation of the ileocecal sphincter, Extrinsic ANS, gastrin



Feedback

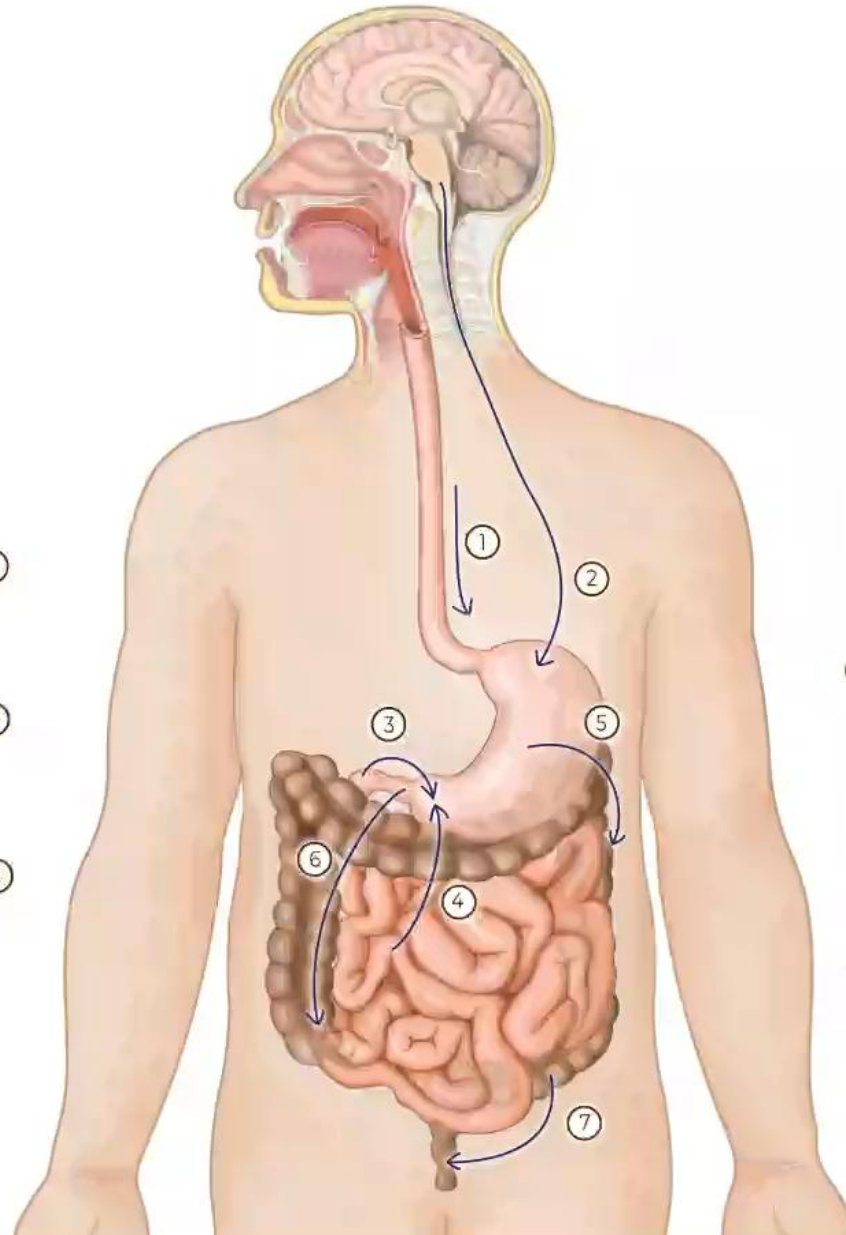
- Enterogastric/ ileogastric– distension and irritation of small intestine results in suppression of secretion and motor activity in the stomach
- Ileal brake – lipid in intestines slows gastric motility
- Rectocolic – colonic peristalsis due to stimulation of rectum

GI reflex overview

Enterogastric inhibitory reflex ③
Stretching of the duodenum
leading to reduced gastric motility

Ileal brake ④
Nutrients in the ileum
delay gastric emptying

Gastroileal reflex ⑥
Stomach and duodenum signal
increased motility in ileum and
opens ileocecal valve



- ① **Receptive relaxation**
Dilation of the stomach fundus when
food passes down the esophagus
- ② **Gastric accommodation**
Subsequent relaxation of the stomach
antrum following receptive relaxation
- ⑤ **Gastrocolic reflex**
Stomach and duodenum signal
increased motility in colon
- ⑦ **Rectoanal inhibitory**
Relaxation of the anal sphincter in
response to distention of the rectum

The role of sphincters

- tonic contractions via specialized circular muscle
- high resting pressures
- relax in response to proximal stimuli
- contract in response to distal stimuli
- regulate anterograde and retrograde flow
- controlled by intrinsic properties of the smooth muscle and neuro-humoral factors

Sphincters	Segments separated
Upper esophageal UES	Pharynx/upper part of esophagus
Lower esophageal LES	Esophagus/stomach
Pyloric	Stomach/duodenum
Sphincter of Oddi	Common duct/duodenum
Ileocecal	Ileum/cecum
Anal- internal & external	Rectum/external environment

Defecation

Internal and External Anal Sphincters

Involved in the Defecation Reflex

Rectum empty: internal sphincter tonically contracted

Rectum fills with feces → rectosphincteric reflex

↓
Internal sphincter relaxes + Urge to defecate

Suppressed by
voluntary contraction
of the external
sphincter

Internal
sphincter regains
tone and rectum
accommodates
to the distension

Rectal wall
relaxes –
pressure
returns to
normal

Process
starts again
and rectum
can store
large amount
of feces

When
appropriate
defecation
takes
place.

- 150 gm excreted each day
- ~100 mls water, 50g solids
- Rectum fills with fecal material, rectum contracts, internal anal sphincter relaxes (rectosphincteric reflex)
- Urge to defecate is felt when rectum is 25% filled
- External anal sphincter remains closed
- When convenient:
 - external sphincter relaxed voluntarily
 - rectum contracts
 - Valsalva maneuver increase intra-abdominal pressure

At this point,
you should be able to

- Understand the difference between tonic and phasic contractions
- Understand how we get peristaltic and segmentation reactions
- Know the mechanisms of the various types of motility seen in each section

Next lecture: Secretions I

Self study:

Small intestinal motility increases as food is being mixed in the stomach.
The ileum prepares for digestion and absorption.

As the stomach empties, intestinal feedback adjusts SI motility so that intestinal digestion and absorption are maximized.

PHASE	STIMULUS	MECHANISM	SMALL INTESTINAL MOTILITY
Gastric	Increased gastric motility	Long neural reflexes Gastrin (Gastroileal reflex)	↑ (activity in ileum)
Intestinal	Distension of duodenum	Neural reflexes	↑(strength of segmenting contractions)

Self study

Reflexes that regulate the activity of the colon.

STIMULUS	MECHANISM	RESPONSE
Increased gastric motility	Long neural reflexes Gastrocolic and gastroileal reflex	Mass movements (gastrocolic) Opening of ileocecal sphincter (gastroileal)
Distension of rectum	Long and short neural reflexes Rectoanal reflex	• Contraction of the rectum • Relaxation of the internal anal sphincter (rectoanal reflex) • Voluntary relaxation of the external anal sphincter • ↑ activity in sigmoid colon
Pain, fear Anger, anxiety, hostility	Extrinsic sympathetic and parasympathetic nerves	• ↓ Motility/constipation • ↑ Motility/diarrhea